Discussion: Implications and Community Recommendations

Relationship to Prior Development Directions

Knight, Cordes, and Friedman knight2022fep identified six development directions for systematic Active Inference literature analysis: (1) increased scope of relevant works, (2) richer annotation schemes, (3) integration of manual and artificial contributions, (4) transferable approaches across fields, (5) participation by diverse contributors, and (6) updated analyses tracking the field's evolution. This pipeline directly addresses directions 1, 2, 3, and 6: it scales retrieval to three databases, replaces manual annotation with LLM-driven extraction while preserving human review pathways, and produces a pipeline designed for incremental re-execution as new literature appears. Directions 4 and 5—cross-field transferability and community participation—remain open and are addressed below.

Tactical and Strategic Priorities

Demand Rigorous Reporting Metadata

Papers should systematically report DOIs, ORCIDs, and explicit hypothesis commitments. Submitted preprints should forward-link to their published versions to prevent fragmented citation subgraphs. Our extraction pipeline prioritizes the DOI as the canonical identifier; failing that, deduplication cascades to arXiv IDs, Semantic Scholar IDs, and OpenAlex IDs. Broad DOI adoption would resolve the cross-source mismatch problem, enabling higher-resolution evidence mapping.

Deploy Open Knowledge Graph Infrastructure

We advocate the deployment of a federated nanopublication server architecture to house community-contributed assertions, enabling a continuously updated living literature review that incorporates new findings as they are published. The release of nanopub-js v0.1.0 kuhn2026nanopubjs makes browser-based creation and querying of nanopublications practical, enabling researchers to contribute assertions directly from web interfaces without requiring command-line tools or Python environments. Integrating this

Empirical and Theoretical Imperatives

Architect Unified Performance Benchmarks

The computational tools domain (B) lacks standardized performance benchmarks for direct comparison against deep reinforcement learning architectures. Establishing baseline metrics analogous to standard RL environments (e.g., OpenAI Gym) is a prerequisite for transitioning theoretical proposals into applied systems.

Prioritize Empirical Validation

Biology (C5) and Language (C3) have established theoretical frameworks but limited empirical validation. Targeted experiments designed to test specific FEP-derived predictions—such as demonstrating morphogenesis as Bayesian inference or measuring active inference advantages in language tasks—would strengthen the evidence base beyond what further theoretical work alone can achieve.

Living Review Maintenance

The pipeline is designed for continuous operation rather than one-time analysis. Incremental resume capabilities (checkpoint-based assertion extraction, merge-on-add corpus deduplication) enable periodic re-execution as new papers are indexed. We envision a maintenance cycle in which the pipeline is re-run quarterly, with updated hypothesis scores and field statistics published alongside the pipeline release. Community contributors can extend the framework by adding custom hypothesis definitions, alternative keyword taxonomies, or domain-specific extraction prompts—all configurable via the YAML configuration file without modifying source code. A complementary long-term trajectory is toward RAG-enabled access: integrating the nanopublication knowledge graph into a Retrieval-Augmented Generation architecture fan2024survey would enable natural-language querying of the evidence base, making quantitative literature synthesis accessible to researchers without programming expertise.

Open Questions

This meta-analysis surfaces questions warranting dedicated investigation:

- **Classifier calibration:** What proportion of A1 papers would be reclassified under embedding-based or expert-annotated schemes?
- **Scoring sensitivity:** How sensitive are hypothesis scores to the choice of weighting function? Would square-root or linear weights qualitatively change the evidence landscape?
- **Model sensitivity:** How much do hypothesis scores vary across different LLM models? Are some hypotheses more robust to model choice than others?
- **Domain boundaries:** Do domain boundaries stabilize as the field matures, or continue to shift? Is the 8-category (A/B/C) taxonomy optimal?
- **Cross-hypothesis evidence:** When a neuroscience (C1) paper supports predictive coding, does this constitute evidence for scalability? How should cross-hypothesis evidence be handled?